

Balance by Gate of Steiner

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Problem Statement: Stress & Workload Manager

Video Presentation: <https://www.youtube.com/watch?v=DV6PFbYdbk8>

Presentation Slides:

https://docs.google.com/presentation/d/1_ctq2F96NrVD1dD9PSV1PLKwfyl83E4G-brSaejZjLE/edit?usp=sharing

1. Project Overview

The Problem

University students often manage assignments, examinations, part-time shifts, family responsibilities, club activities, group work and personal rest within the same limited period. The core problem occurs when the time and energy required by these commitments exceed the student's actual capacity. Contributing causes include overlapping deadlines, incomplete task estimates, unclear flexibility, unprotected sleep or recovery limits, and shared commitments that cannot be changed without another person's agreement.

Students may recognise that they are overloaded but still cannot answer three practical questions: What must remain protected? What can be adjusted? What consequences will each adjustment create? Moving one task may protect tonight's sleep while increasing tomorrow's workload, creating deadline risk or affecting teammates. Students are the primary stakeholders; secondary stakeholders include teammates, family members, club members and university support services affected by or supporting the student's decisions.

Existing solutions address parts of this problem. Todoist supports task organisation, replanning and collaboration, while Finch supports self-care goals and gamified engagement. Balance explores how protected commitments, adjustment consequences, agreement requirements and recovery time can work together in one workload decision flow. A detailed comparison appears in Section 4.

Our Solution

Balance is a stress and workload management application that helps university students create a realistic plan when every commitment cannot fit within their available time. It follows a Detect -> Decide -> Recover -> Reflect process: the system identifies overload, protects non-negotiable commitments, and compares possible adjustments before the student confirms any change. Each option explains its effects on deadlines, future workload, rest and collaboration. Once a confirmed plan creates genuine free time, Balance helps the student protect that time for optional recovery and later reflect on recurring workload patterns.

Feature Set

Practical feature	RPG label	Purpose
Workload Overview	World Status	Shows available and planned time, five workload dimensions and an illustrative cumulative workload trend; the model remains to be validated.
Optional Check-in	Status Appraisal	Records optional sleep and self-reported energy. Missing input remains Unknown.
Tasks and Protected Commitments	Quest Board and Sacred Contracts	Records duration, deadline, flexibility (Fixed / Flexible / Needs Agreement), protection status and a separate optional-commitment flag.
Overload Alert	Calamity Alert	Shows the difference between genuinely available time and planned task time.
Plan Comparison	War Council	Compares valid adjustments and explains deadline, future-load, rest and collaboration effects.
Recovery Nudge	Sanctuary	Offers optional recovery only after a confirmed plan creates genuine free time.
Weekly Reflection	Journey	Summarises private weekly patterns, protected time and conflicts handled early.
Sustainable Achievements	Journey Milestones	Recognises safe planning without leaderboards, streak penalties or rewards for overwork.

2. Ideation & Process

2.1 Ideas We Considered

Direction	Decision	Reason
Debt-based Workload Model + Load Balancer + Recovery Nudge	Chosen	Connects cumulative workload detection, a concrete planning decision and protected recovery time.
RPG framing	Refined and retained	Makes the experience memorable while keeping practical labels visible and avoiding a complex game economy.
Journey and Sustainable Achievements	Added	Responds to mentor feedback on retention by rewarding safe planning and private reflection without streak pressure.
Negotiation AI	Refined	Changed from generated external messages into user-confirmed plan comparison and a Needs Agreement state.
Energy Currency	Dropped	Too difficult to explain credibly in a 3-5 minute demo and shifted attention toward a game economy.
Peer Comparison	Dropped	Added privacy and multi-user complexity; comparison could also increase anxiety.

2.2 Ideation Boards

Early Problem and Solution Exploration

These three original handwritten pages record early problem framing, proposed solutions and unresolved questions in Chinese. The English summaries below describe the legible content rather than provide a word-for-word transcription. They document design exploration, not user interview findings or validated outcomes. Their order follows the supplied scan; it does not establish separate dated iterations.

Early draft A (7/9/2026) — Practical constraints and continued use

学生常见问题困扰

- 没实际用处
- 没有具体和针对性方案
- 无法解
- 下载,但不一定用或一直用 一要有-种方案让用户有动力使用
- 游戏化每日任务?
- 有成就感

- 隐私问题

- 安全问题

- 建议需走进生活

有手
眼

- 可能没精力和时间

- 很多压力来源必要的活动如 assignment 或工作
- 不能假设所有的任务都减少

需要跟据实际场景来考虑
如 part time

- 核心问题:如果学生已经 Over, 是什么导致

Early draft A. Original Chinese handwritten notes

English summary: The notes consider why students may not continue using an app and explore gamification, daily tasks and a sense of achievement. They also raise privacy and security concerns. A key observation is that stress can arise from necessary commitments, including assignments and part-time work: the system cannot assume every task can be reduced. Advice should fit real circumstances.

Connection to the current design: protected commitments and consequence-aware trade-offs address the concern that necessary responsibilities cannot simply disappear. Journey Milestones retain the achievement idea while avoiding compulsory daily tasks and streak penalties.

Early draft B (7/9/2026) — Proposed responses to student needs

1. 几个重要任务同时的问题

— 难以先处理哪件

方案1: 未来推导, 显示出如果先完成哪个任务时的成果, 如: 未来因此改变的 task list, 因此变好或变坏。

2) 有些事情难以自由调整

比如工作 家庭上

方案1: 可选优先级, 以这种为例, 系统检测到后, 不会调整其时间, 反而会有这种排定时间。

3) 学业和经济上的压力太大, 难以解决

方案1: 对于学业

没想法

4) 需要个性的解决方案, 比如根据当时的压力和压力的任务, 来显示附带的方案。

5) 加重安全管理

6) 访问详细的个人信息, 并以推送符合个人情境

推送内容

什么类型的推送

Early draft B. Original Chinese handwritten notes

English summary: The notes explore showing the effects of completing tasks, protecting work and family time, providing personalised suggestions based on available time and stressful tasks, and strengthening security. They acknowledge academic and financial pressures that may be difficult for the app to resolve. Collecting profile information such as year of study and field of study is considered as a possible basis for personalisation.

Connection to the current design: fixed and protected commitments, capacity checks and plan comparison make these proposals more specific. Profile-based personalisation remains an early proposal rather than a committed feature. The current scope focuses on workload decisions and does not claim to resolve financial or academic pressures themselves.

Early draft C (7/9/2026) — Scheduling rules and unresolved questions

每日资料输入,使其精简,还输入昨日睡眠时长。

对于超载转移,

一会给每日的量定一个上限,会转移至适当的尺寸但在期限内
如果遇到都满的时候,要求用户取舍。

假设学生拒绝或没时间配合的话,

一没想法

压力分数所计算的压力指数可信问题,
那就严格化五围扣减的规则,使其严谨

缺少更新的问题系,就以用户再次回归,显示一个比仔细的问题,以用于
判断用户的状态,以此再计算压力值。

Early draft C. Original Chinese handwritten notes

English summary: The notes propose keeping daily input short, including the previous night's sleep duration, setting a daily task limit and moving work to a suitable day within its deadline. They recognise cases requiring user input. They explicitly leave open what happens if the student rejects a schedule change, question the credibility of a calculated stress score, and consider asking returning users for updated information when check-ins are missing.

Connection to the current design: changes require confirmation; rejection preserves the current plan; missing information remains Unknown or Needs Review; and pending agreement does not count as freed time. The credibility question remains relevant: cumulative workload rules require definition and validation, and the prototype does not claim a clinically valid stress score.

The following mind map and problem tree organise the problem and design relationships more explicitly. The original notes above are retained as evidence of the earlier questions and proposals, including ideas that were later narrowed or left unresolved.

Design Evolution

Retrospective Design Evolution

Earlier normal stress tracking	→ Revised design More detail mode Detect, Decide, Recover, Reflect	→ Reason to make student can act on overload and Review patterns, no just normal stress tracking
2) Automatic Rescheduling	detail the process compare, confirm, reject, Undo	keep students in control and show consequences.
3) Move task to fit	Protect Sacred Contracts	Preserve sleep, work and family duties, show No Feasible Plan when necessary
4) Move group meeting	Needs Agreement	keep shared commitment unchanged until agreement is recorded
5) Recovery task	Protect, optional recovery time	student may leave the slot unplanned with losing it.
6) Recovery lowers calamity	Separate records	Protect time, activity completion and report energy are different.
7) Complex RPG word	Simple RPG labels with practical names	keep the experience engaging and understandable.
8) Daily game pressure	Journey and sustainable milestones	support reflection without streak penalties or rewards for overwork.
9) Single-day rescheduling	check the destination day	consider deadlines, future capacity and collaboration.

Mindmap

Draft version

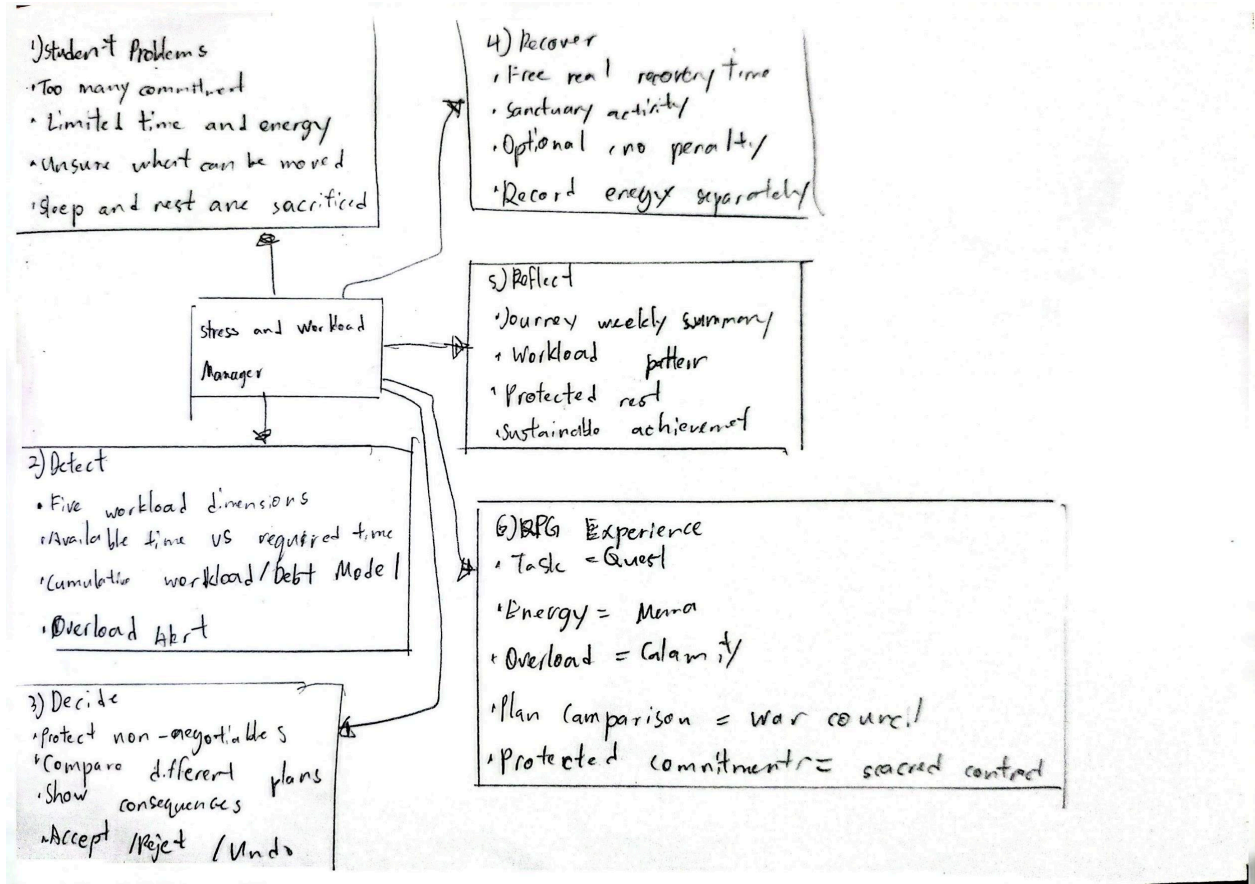


Figure 1. Original hand-drawn mindmap

Final version

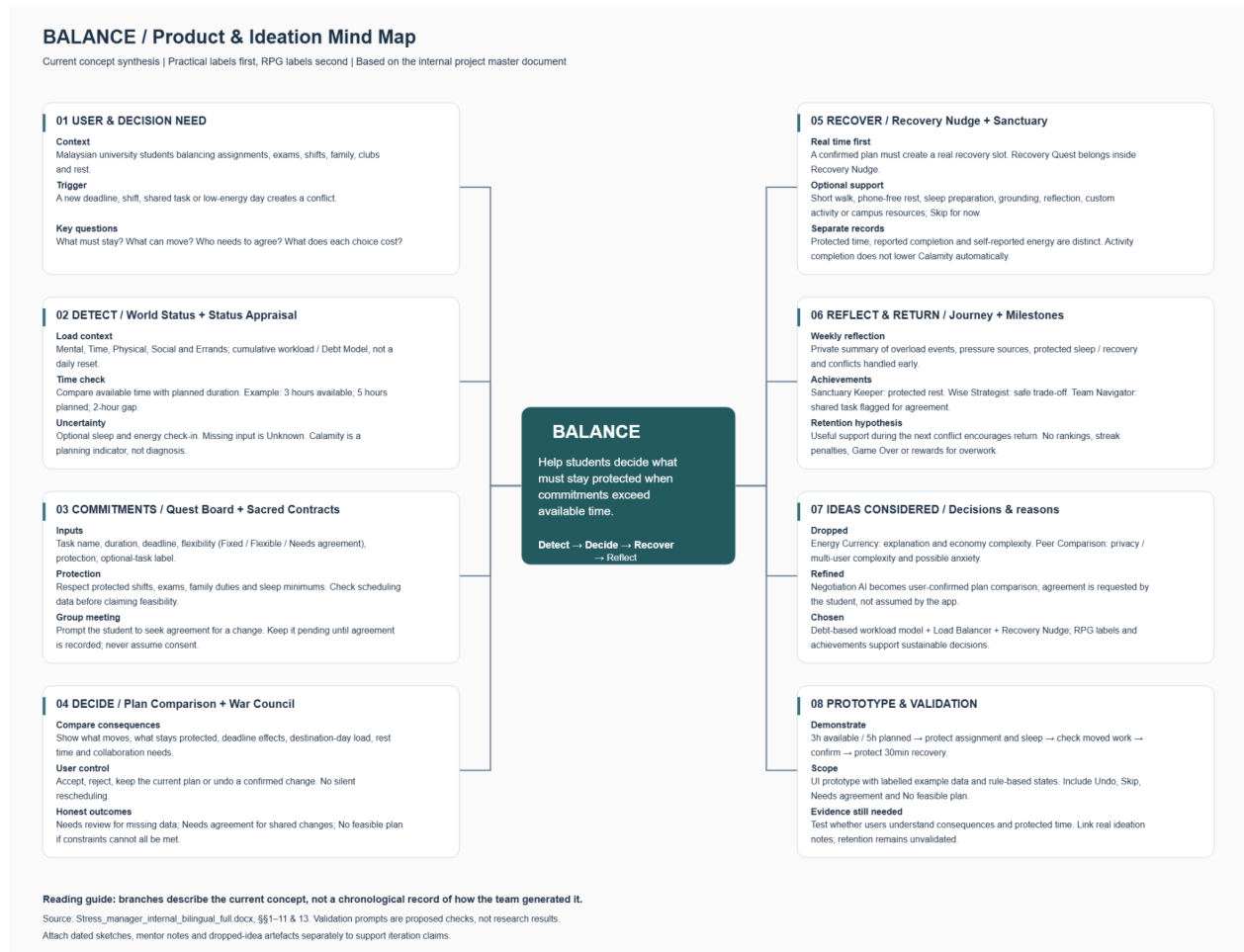


Figure 2. Refined mind map connecting the user problem, core product flow, considered ideas, trust rules and prototype validation needs.

The mind map connects student workload conflicts to the Detect → Decide → Recover → Reflect journey. It links capacity awareness to protected commitments and consequence-aware plan comparison, then connects genuine free time to optional recovery and private reflection. Practical feature names accompany RPG labels to explain their purpose.

Problem Tree

Draft version

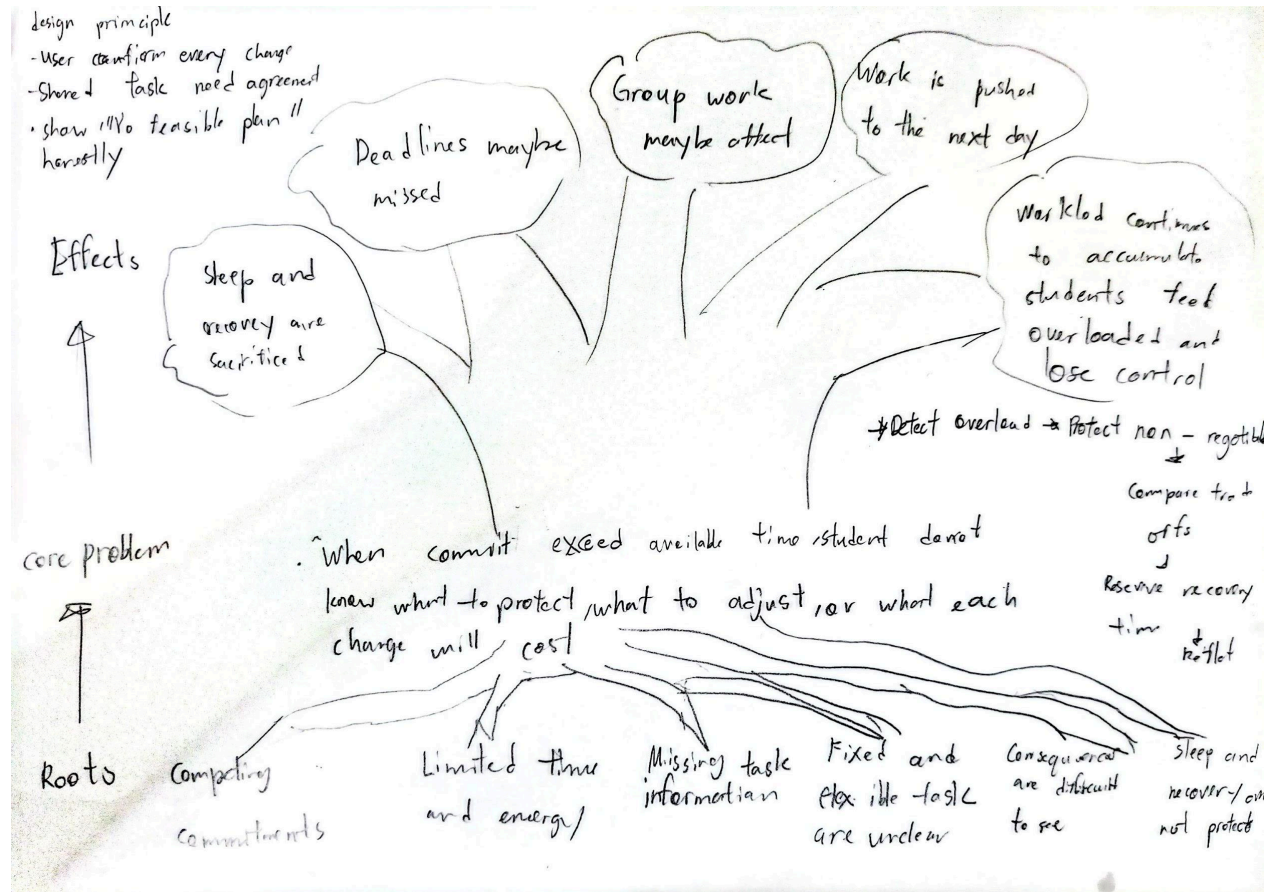


Figure 3. Original hand-drawn problem tree showing causes, the core decision problem, possible effects and early design principles.

Final version

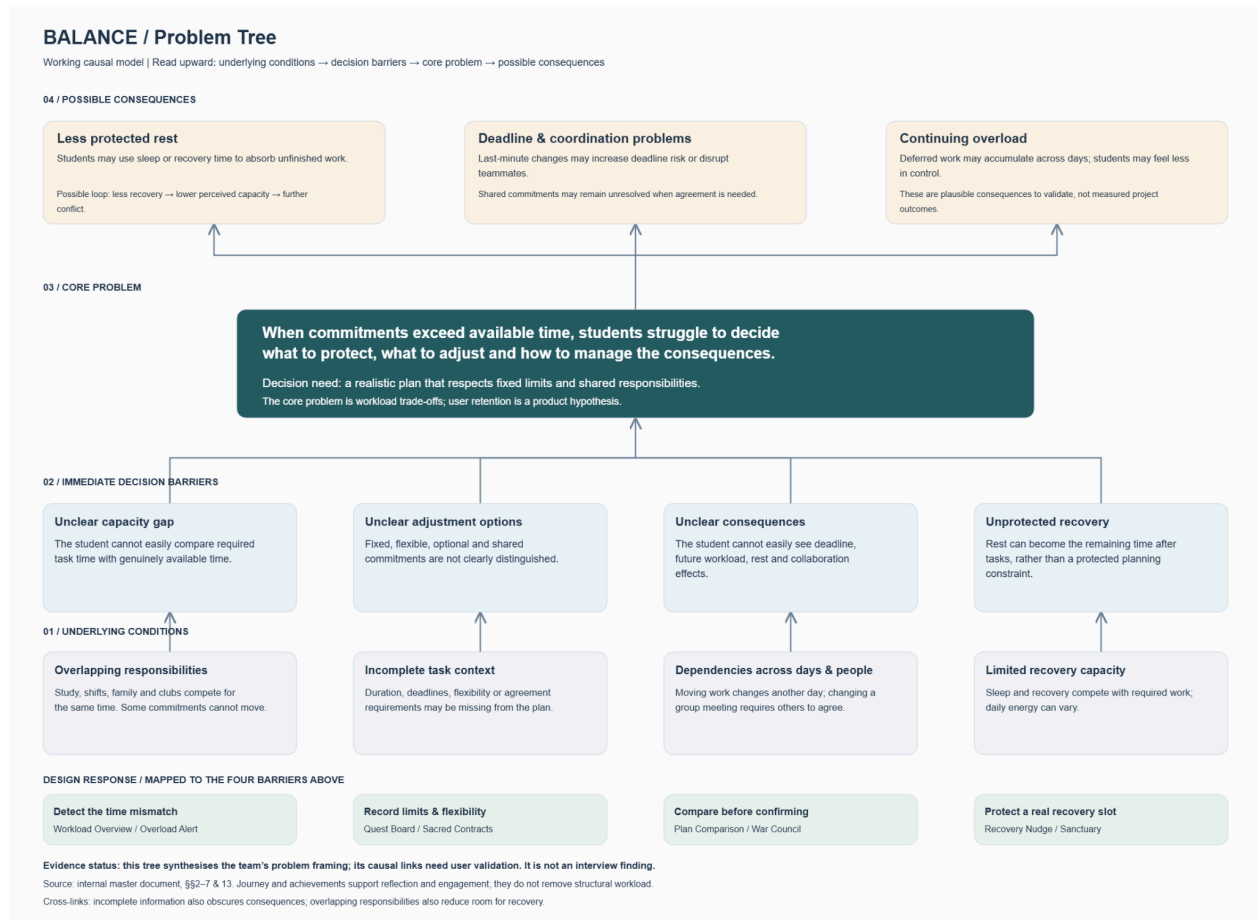


Figure 4. Refined working causal model. Its causal links remain hypotheses to validate and are not presented as interview findings.

The problem tree links overlapping responsibilities, incomplete task information and dependencies across days and people to four decision barriers: an unclear capacity gap, uncertain adjustment options, hidden consequences and unprotected recovery time. These barriers informed Workload Overview, Quest Board and Sacred Contracts, War Council, and Sanctuary. The causal links are working hypotheses to validate, rather than confirmed interview findings.

2.3 Mentor Consultation

Date	Mentor	Feedback received	What changed
6 Sep 2026	Lim Zi Yang	Develop a distinctive response beyond reproducing the problem statement. Consider multi-device use, privacy and security, and avoid relying on AI for the team's original brainstorming.	Added protected commitments and consequence-aware trade-offs; removed peer comparison; made self-reporting optional; limited the prototype to rule-based example data and avoided medical claims.
8 Sep 2026	Lim Zi Yang	Define a specific use case, examine existing stress and workload apps, and decide how physical or mental wellbeing and gamification support that use case.	Defined the 3-hours-available versus 5-hours-planned scenario. Recovery now appears only after real capacity is freed, while RPG labels support the planning flow.
11 Sep 2026	Lim Zi Yang	Reconsider how group meetings are changed. Address retention, progress reflection, expanded Sanctuary support and game elements.	Added Needs Agreement for shared commitments; Journey as a private weekly reflection; sustainable planning achievements; and broader optional Sanctuary activities. Workload trade-offs remain the core user problem; retention features support continued use.

3. Design & Prototype

UI Prototype:

<https://www.figma.com/design/oLcPugPWY5bqWTPPxMYtm/Balance-%E2%80%94-RPG-Workload-Manager?node-id=0-1&t=TJToq2FSosvTu5Mh-1>

The current Figma prototype presents eight ordered mobile UI screens illustrating a university student with three hours available tonight and five hours of planned tasks. These are static reference screens showing the intended decision journey. Clickable transitions, working confirmation and undo, and rule-based calculations are planned for implementation.

The experience follows the **Detect** → **Decide** → **Recover** → **Reflect** journey:

Detect — The intended interface combines a calculated time-capacity gap with self-reported information across Mental, Time, Physical, Social and Errands. Task flexibility is Fixed, Flexible or Needs Agreement; protection status and whether a commitment is optional are recorded separately. The example shows a two-hour capacity gap.

Decide — Proposed adjustments explain what moves, what stays protected, deadline effects and destination-day capacity before confirmation. A shared commitment remains unchanged until the student records that the required agreement has been obtained; pending agreement does not count as freed time.

Recover — In the corrected scenario, moving 90 minutes of reading and 60 minutes of draft work removes 150 minutes from tonight. The remaining 150 minutes of tasks fit within 180 available minutes, leaving 30 minutes for optional recovery or unstructured rest.

Reflect — The weekly screen illustrates private patterns using example data. Working history tracking and milestone rules are planned; the current screens do not demonstrate measured retention or wellbeing outcomes.

The intended naming convention pairs practical feature names with RPG labels: Tasks / Quest Board, Protected Commitments / Sacred Contracts, Plan Comparison / War Council, and Recovery Nudge / Sanctuary.

Key Screen Flow

Step	Practical feature / RPG label	Intended interaction
1	Optional Check-in / Status Appraisal	The student may record current energy and rest information. These optional readings inform the status view but do not automatically change the schedule.
2	Workload Overview / World Status	The interface compares 3 available hours with 5 planned hours, states the 2-hour capacity gap, and displays Mental, Time, Physical, Social and Errands separately.
3	Commitment Map / Quest Board	Protected sleep and family commitments are separated from fixed and flexible tasks, so essential commitments cannot be moved merely to make the plan appear feasible.
4	Plan Comparison / War Council	The reference screen illustrates proposed adjustments. Implementation must validate duration, deadlines, destination-day capacity, protection and agreement before labelling an option feasible.
5	Confirmed Plan / Plan Updated	The intended confirmation state lists the two moved task portions and protected commitments. Undo will restore both changes and the prior recovery reservation.
6	Recovery Nudge / Sanctuary	The corrected scenario leaves 30 minutes after moving 150 minutes of tasks from a 300-minute plan into a 180-minute window. The student can reserve this time or leave it unplanned.
7	Weekly Summary / Weekly Reflection	A private weekly view presents workload patterns, protected recovery time and sleep protection without comparing the student with other users.
8	Planning Achievements / Journey Milestones	Cumulative milestones recognise sustainable decisions such as protecting sleep, keeping recovery time clear and rebalancing early, without streak penalties or peer rankings.

Core Interaction Principles

- User control: no task or commitment moves without confirmation.
- Protected constraints: sleep, family responsibilities and other non-negotiable commitments cannot be sacrificed simply to make the schedule appear feasible.
- Visible consequences: every proposed adjustment explains its cost before the student decides.
- Optional recovery: Sanctuary protects genuine free time without turning recovery into another compulsory task.
- Clear terminology: practical feature names appear together with their RPG equivalents.
- Honest feedback: calculations, self-reports and example data are distinguished. The product makes no medical, diagnostic or therapeutic claims.

Selected UI Reference Screens



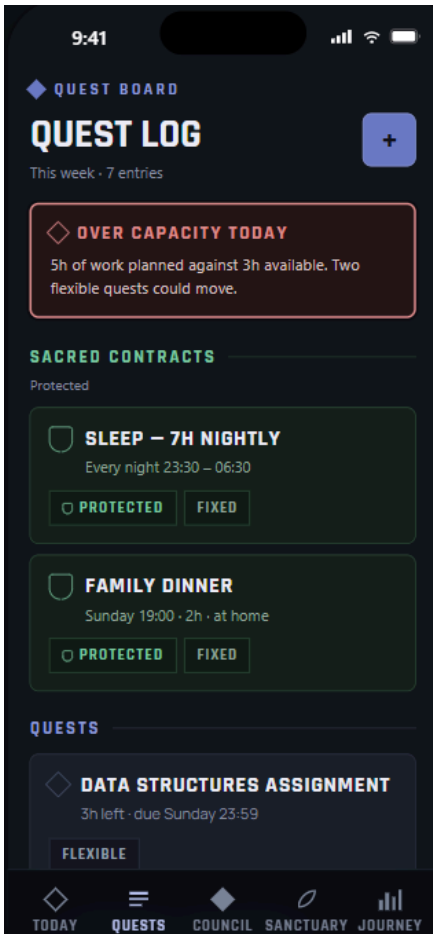
1. Optional Check-in / Status Appraisal

Optional energy and rest inputs inform suggestions without changing the schedule.



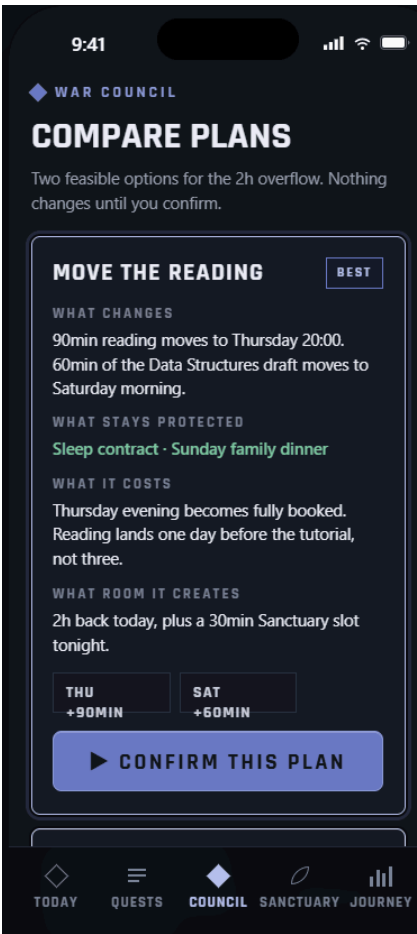
2. Workload Overview / World Status

The screen compares three available hours with five planned hours and shows the two-hour gap.



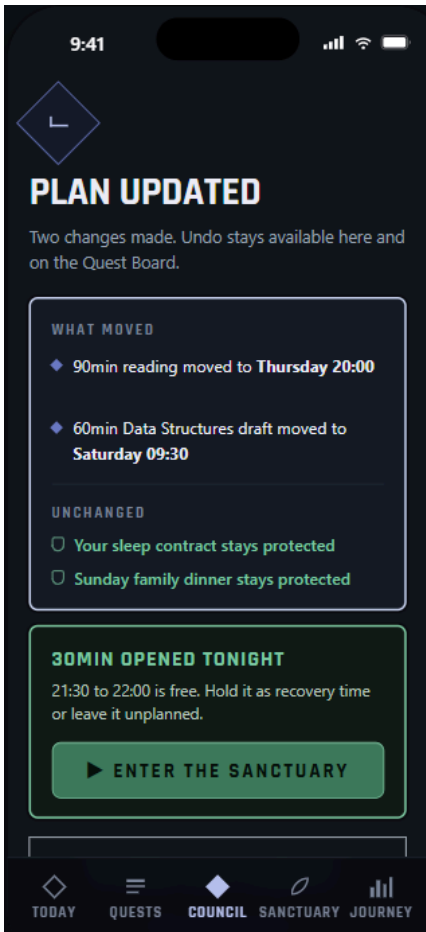
3. Commitment Map / Quest Board

Protected commitments remain fixed. The Data Structures task is due Sunday at 23:59.



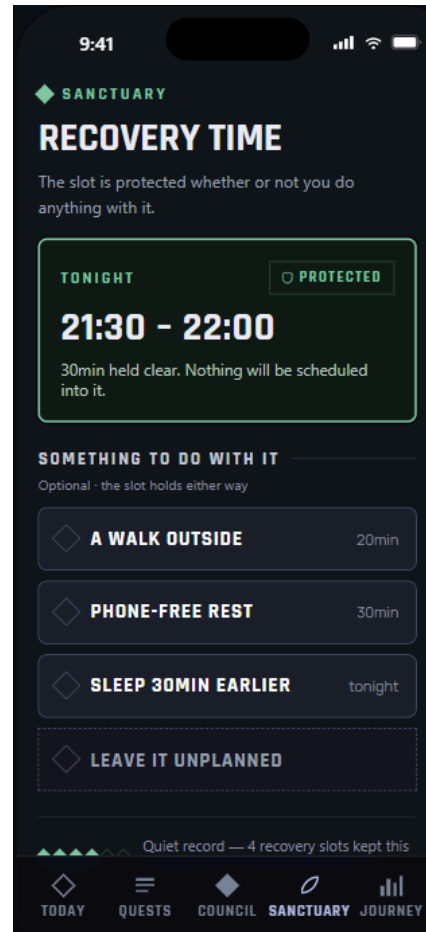
4. Plan Comparison / War Council

The proposed plan states what moves, what stays protected and what each choice costs.



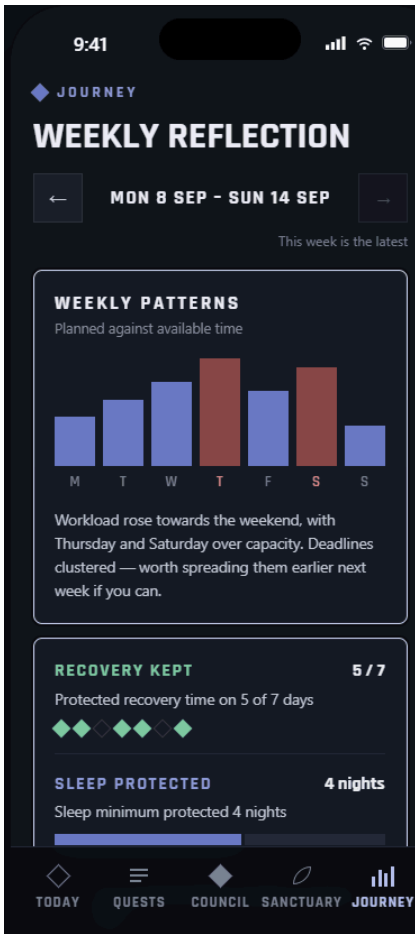
5. Confirmed Plan / Plan Updated

Moving 150 minutes of work leaves 30 minutes tonight; Undo remains available.



6. Recovery Nudge / Sanctuary

The corrected protected period runs from 21:30 to 22:00 and may remain unplanned.



7. Weekly Summary / Weekly Reflection

Illustrative private data shows patterns without comparing the student with other users.



8. Planning Achievements / Journey Milestones

Milestones recognise sustainable planning without streak penalties or peer rankings.

4. What Makes It Different

Distinctive mechanism	Why it matters
Non-negotiable Commitments	Students protect a shift, exam, family duty, shared task or sleep minimum. The plan cannot sacrifice these simply to appear successful.
Consequence-aware Trade-offs	Balance previews what changes, what remains protected, destination-day capacity, deadline effects, rest effects and agreement needs before confirmation.
Recovery Follows Real Capacity	Recovery appears only after a confirmed change creates time. Protected time, reported completion and self-reported energy remain separate.
Honest Uncertainty	Missing information becomes Needs Review; shared changes become Needs Agreement; impossible constraints return No Feasible Plan.
Light RPG Framing	Every RPG label is paired with a practical name. Milestones reward sustainable decisions without leaderboards, streak penalties or Game Over states.

Comparison with Existing Solutions

We reviewed official feature descriptions from Todoist and Finch to identify where Balance could add value.

Existing solution	Documented capabilities	Balance's proposed distinction
Todoist	Task priorities, schedule replanning, shared projects, task assignments and productivity tracking.	Balance proposes a guided comparison that checks protected commitments, destination-day capacity and agreement requirements

		before a schedule change is confirmed.
Finch	Self-care goals, daily Quests, Streaks and weekly milestones for self-care areas.	Balance connects optional recovery to time actually released by a workload decision. A recovery slot may remain unplanned, and activity completion does not automatically imply improved energy.

Sources: [Todoist official features](#) and [Finch official features](#).

Our differentiation is the combined decision flow. For a student with three available hours and five hours of work, Balance is designed to identify what must stay protected, explain which changes are possible, check their consequences, and reserve any remaining time for optional recovery. A shared commitment awaiting agreement does not count as freed time.

RPG labels and achievements support this flow; they are not claimed as independently original features. Balance’s proposed milestones focus on sustainable planning, such as protecting sleep and addressing conflicts early.

This comparison reflects the official pages reviewed, rather than exhaustive product testing. It does not establish that competitors cannot support similar workflows. Balance currently demonstrates the proposed approach through static prototype screens; its decision rules and user benefits still require implementation and validation.

5. Technical Architecture & Feasibility

Tech stack

Layer	Planned choice	Reason and constraint
Prototype	Figma static screen flow	Eight ordered reference screens illustrate intended states. Clickable navigation, confirmation, undo and calculations are not implemented in this deliverable.
Product frontend	React, TypeScript and Tailwind CSS	Supports a responsive mobile-first web interface with one codebase.
Decision engine	Client-side rule-based checks	Checks duration, deadlines, destination capacity, protected commitments and agreement status without machine learning.
MVP data	Seeded example data / in-memory browser state	Planned local demo state resets on refresh. Persistence will be added through a repository adapter after the flow is validated.
Stretch data and authentication	Supabase PostgreSQL and Supabase Auth	Adds per-user persistence after the interaction flow is stable; row-level access policies would be required.
Hosting	Vercel	Provides a straightforward deployment path for the React application.

Architecture Patterns

The planned product uses a modular monolith with layered responsibilities. MVVM separates views from presentation state; application use cases coordinate Detect, Decide, Recover and Reflect; the domain core contains workload, constraint, trade-off and recovery rules; and Repository interfaces isolate storage. The Trade-off Strategy can evolve without changing the UI, while explicit UI states keep confirmation, agreement and failure conditions visible.

The architecture follows three SDA principles used by the team: Separation of concerns, Dependency direction toward core policies, and Abstraction through application-owned ports. Calendar, notification, campus-resource and cloud adapters remain optional until implemented and tested.

Confirmation and undo behaviour

The planned Confirm Plan use case revalidates durations, deadlines, destination-day capacity, protected commitments and agreement status. If any required information is missing or a constraint fails, it returns Needs Review, Needs Agreement or No Feasible Plan without changing the schedule. A valid confirmation stores the previous plan, applies the selected changes and updates the UI. Undo restores the previous task placements and recovery reservation together.

Workload model and data boundaries

Workload Model and Data Boundaries

Time-capacity calculation. Balance defines the scheduling gap as:

`Overload minutes = max(0, planned task minutes - available minutes)`

Available minutes exclude fixed commitments and protected time. Commitments already excluded from availability must not be counted again as planned tasks. For example, 300 minutes of planned work within 180 available minutes creates a 120-minute gap.

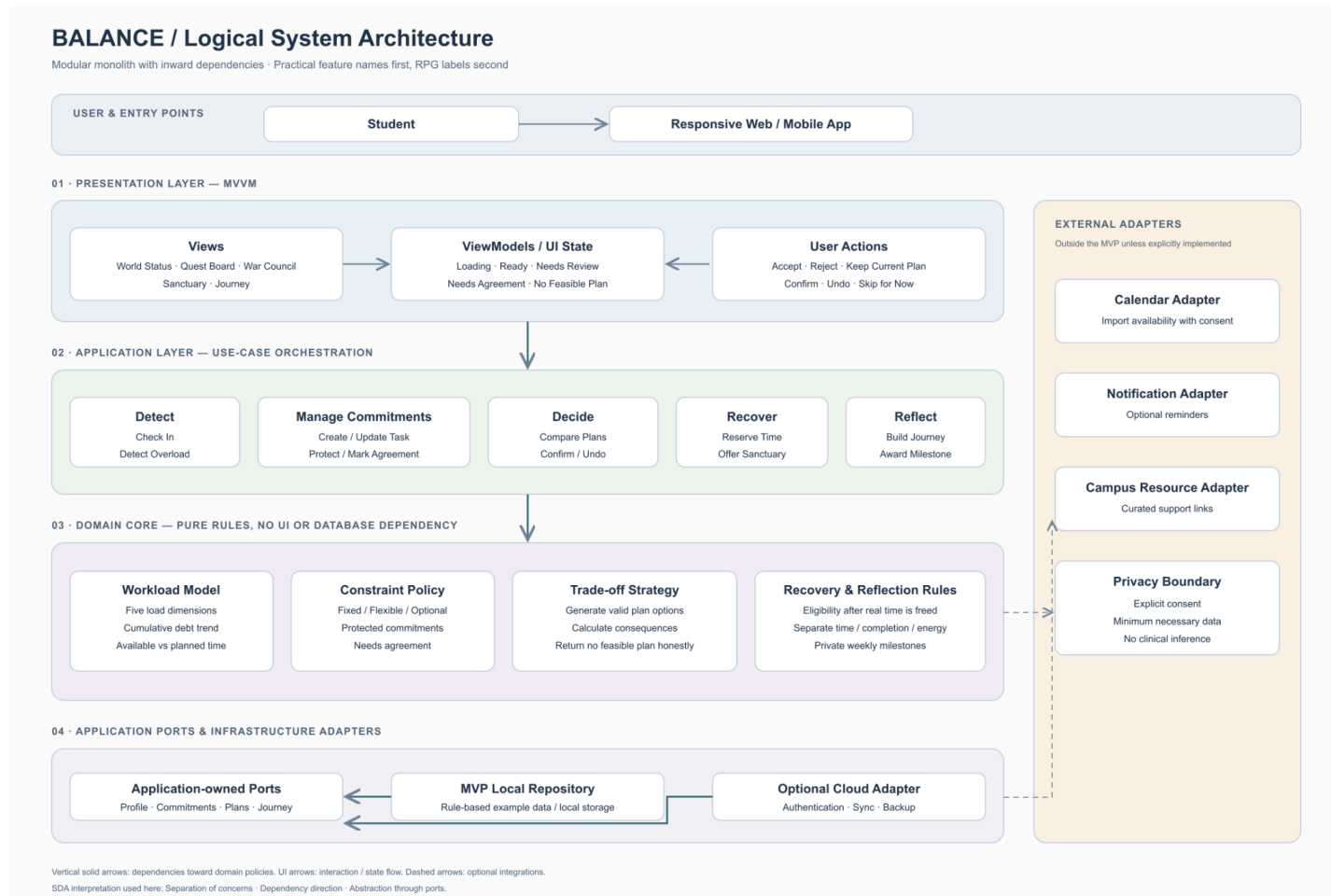
Debt-based Workload Model. The proposed model represents workload pressure that can carry across days rather than automatically resetting each morning. The prototype illustrates this concept using example data. Its accumulation, carry-over and update rules have not yet been validated, so Calamity is not presented as a measured stress score or a medical assessment. Mental, Physical, Social and Errands use labelled self-reports; missing information remains Unknown.

Moving a task changes when its workload is scheduled; it does not remove that workload. Likewise, reserving or completing a recovery activity does not automatically reduce Calamity. Protected recovery time, activity completion and self-reported energy are recorded separately.

Planned decision rules. Before confirming a plan, the implementation will check task duration, deadlines, destination-day capacity, protected commitments and agreement status. Missing information returns Needs Review; an unapproved shared-task change remains Needs Agreement; and a plan that cannot satisfy the constraints returns No Feasible Plan. These rules are specified in the design but are not yet executed by the static prototype.

Validation plan. We will test the scheduling rules using normal, overloaded, missing-information and pending-agreement scenarios. Student walkthroughs will assess whether users understand the proposed changes and their consequences. The cumulative workload model will require separate evaluation against repeated self-reports before its scores are used to guide decisions.

System Architecture



Build plan & scope

The committed build focuses on one complete local decision journey. The current deliverable is a static Figma screen flow. The planned first implementation uses seeded example data and in-memory browser state; changes reset on refresh until a persistence adapter is implemented. Authentication and per-user cloud storage remain stretch work.

Phase	Deliverable	Scope
1. Foundation	Set up the responsive app shell, navigation, shared components and consistent practical/RPG terminology.	Committed
2. Detect	Build task entry, optional check-in, protected commitments, five load dimensions and the capacity-gap state.	Committed
3. Decide	Implement rule-based plan options, consequence summaries, Needs Review, Needs Agreement, No Feasible Plan, Confirm and Undo.	Committed
4. Recover	Reserve genuine free time and provide optional Sanctuary activities with Skip for Now.	Committed
5. Reflect	Create the Journey weekly summary and non-repeatable sustainable planning milestones.	Committed
6. Validate and polish	Test the 3-hour versus 5-hour scenario for comprehension, navigation and consistency; fix unclear copy and states.	Committed

7. Persistence	Add Supabase authentication and per-user storage only after the complete local flow works reliably.	Stretch
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Out of Scope for This Build

Machine learning, burnout diagnosis or prediction, therapy claims, automatic messages to lecturers or teammates, live calendar integration, push notifications, social comparison, multiplayer systems, combat mechanics and an equipment economy are outside this build.

References

Features | Todoist. (n.d.).

<https://www.todoist.com/features>

Finch. Finch Features.

<https://help.finchcare.com/hc/en-us/categories/37934152903309-Finch-Features>